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SPPIRO: a Soft, Power-autonomous, Proprioceptive In-pipe Robot for adaptive inspection in complex pipeline environments

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Subterranean and industrial pipelines are critical infrastructure for transporting water, oil, and gas, yet their internal complexity—featuring varying diameters, sharp turns, and multi-directional junctions—poses significant challenges for conventional inspection robots. We present SPPIRO, a fully untethered soft robot designed for adaptive pipeline inspection in such environments. SPPIRO integrates compact, custom-designed PCBs for onboard power, sensing, and control, enabling fully untethered operation. Its anchoring modules use motor-driven six-bar linkages with 3D-printed, honeycomb-inspired soft wheels, allowing adaptation to pipeline diameters ranging from 140 to 200 mm. The propulsion module consists of three self-sensing pneumatic Kresling origami actuators arranged at 120°, providing omnidirectional steering and real-time body shape estimation. Using a Piecewise Constant Curvature (PCC) model, SPPIRO reconstructs both its own configuration and the surrounding pipe geometry during motion. Liquid-metal-based capacitive sensors embedded in the feet enable contact-aware anchoring, even in opaque pipelines. Experiments demonstrate SPPIRO's ability to traverse vertical and horizontal segments, 90° turns, and T-junctions, achieving speeds up to 9.5 mm/s—outperforming prior untethered soft in-pipe robots. These results position SPPIRO as a promising platform for autonomous, geometry-aware pipeline inspection.

Oil and natural gas remain primary energy sources powering modern transportation, electricity generation, and industrial operations. The distribution of these resources relies heavily on vast pipeline networks that transport crude oil, refined products, and natural gas. The networks often span long distances, feature varying diameters, and include complex geometries- ranging from sharp 90° turns to T-junctions. Usually buried underground or deep sea, pipeline inspection becomes challenging and dangerous with existing human-operated technologies. As a result, deterioration goes undetected until leaks occur, leading to costly economic damage and environmental harm.

To prevent such failures, various robotic solutions have been developed for pipeline interior inspection, including track-based[1], wheeled[2], walking[3], and other locomotion strategies[4–8]. Most of these robots rely on electromagnetic motors and gear-driven mechanisms, making them well-suited for inspecting large-diameter pipelines and executing high-speed maneuvers across long, straight, intercontinental routes. However, when navigating small-diameter pipes—often only a few to tens of centimeters wide—with sharp turns and complex junctions, traditional rigid robots face significant challenges. Their inflexible structures, limited articulation, and insufficient adaptability hinder their ability to maneuver through tight bends and intricate geometries.

In recent years, researchers have explored soft in-pipe robots constructed from compliant materials and actuators. Tang et al.[9] utilized dielectric elastomer actuators (DEAs) to develop a fast-moving, sub-centimeter-scale pipeline-inspection robot, achieving a maximum speed of 1.19 body lengths per second. Zhang et al.[10], along with other groups working on pneumatically actuated in-pipe robots[11–16], have drawn inspiration from earthworm locomotion and demonstrated a variety of robust robots capable of maneuvering within confined spaces. In addition, Li et al.[17] developed a manta-ray-inspired untethered soft robot capable of in-place steering within multidirectional pipe junctions. Mao et al.[18] proposed a magnetically driven soft continuum robot with a cross-sectional diameter of approximately 3–4 mm that can steer and navigate through delicate and complex lumen- and blood-vessel-like tubular structures.

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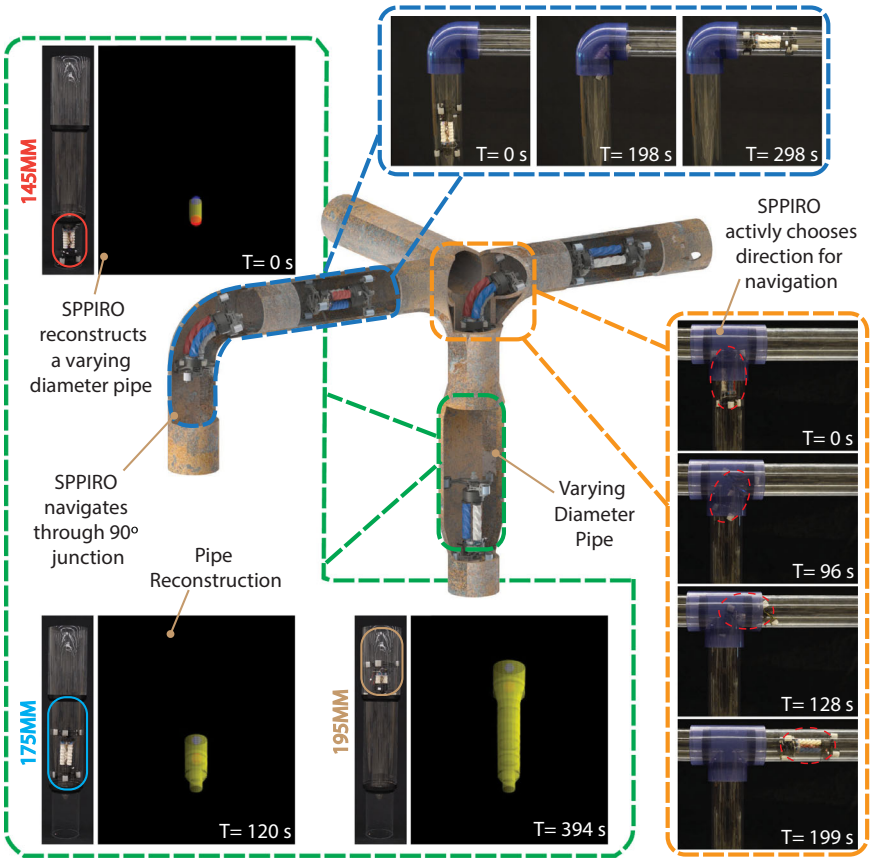


Fig. 1 | Overview of the fully untethered in-pipe robot SPPIRO. SPPIRO is capable of navigating both horizontal and vertical pipelines, performing omnidirectional steering, completing 90° turns (blue box), and traversing T-junctions (orange box). It can also maneuver through pipelines of varying diameters while reconstructing their geometries (green box).

Vine robots[19], which extend from the tip like a growing vine, have attracted significant attention due to their large reachable workspace, compact stowability at the base, compliant locomotion that enables squeezing through obstacles, and scalable design. These features allow deployment in sub-centimeter-scale pipes—such as lumens and blood vessels for medical procedures[20]—as well as sub-meter-scale industrial pipeline systems[21]. Vine robots have also demonstrated steering capabilities in multidirectional pipe junctions: Behnke et al.[22] achieved navigation through T-junctions via robot self-interaction, and Qin et al.[23] designed a steerable tip to direct the eversion path.

Despite these promising advances, gaps remain between current prototypes and real-world deployment. DEA-based robots and most pneumatically driven systems are tethered, restricting their operational range in long-distance inspections. Moreover, these robots typically rely on passive turning through environmental interaction, which limits their effectiveness in pipeline networks containing multidirectional junctions. Magnetically driven robots must operate under externally applied magnetic fields, making them well-suited for miniature tubular environments within the human body but impractical for larger-scale industrial pipelines. Vine robots face inherent challenges in retraction, especially over long deployed lengths, where the thin body is prone to buckle under tail-applied forces. Incorporating steering mechanisms-such as tendons[24], variable-stiffness elements[25], or LCE actuators[26]- adds rigidity that further complicates smooth inversion during retraction. This added stiffness also increases the robot's bending resistance, which can limit steering angles and require higher internal pressure to turn inside complex multidirectional junctions. As a result, few existing soft robotic solutions have been successfully deployed in real-world pipeline networks for inspection, especially in the pipelines at the sub-centimeter to sub-decimeter scale.

To address these challenges, we propose **SPPIRO**, a **S**oft, **P**ower-Autonomous, **P**roprioceptive **I**n-Pipe **RO**bot capable of navigating pipes with varying diameters, sharp turns, and multidirectional junctions under fully untethered operation. SPPIRO has an overall length of 250 mm and a body diameter of 68.5 mm, making it compact enough for confined environments. It employs three 3D-printed pneumatic Kresling origami actuators that achieve large contraction (~70%) and enable active omnidirectional in-pipe steering. Leveraging the unique coupling of twisting and contraction in the Kresling origami structure, the robot can estimate the length of each actuator in real-time using onboard encoders, allowing real-time robot shape reconstruction. To adapt to pipes with varying diameters, two sets of multi-linkage mechanisms, driven by micro motors, provide a radial expansion ratio exceeding twofold. Three 3D-printed compliant wheels with honeycomb structures are integrated into each anchoring mechanism to enhance wall contact and ensure secure anchoring during steering. Additionally, liquid-metal-based soft capacitive sensors embedded in the wheels enable reliable contact detection. Our experiments demonstrate SPPIRO's ability to navigate both upward and downward through pipelines ranging from 145 to 195 mm in diameter with real-time pipe diameter mapping (Fig. 1, green box), perform 90° turns (Fig. 1, blue box), and traverse T-junctions in both directions (Fig. 1, orange box). Equipped with an onboard camera and contact sensors, the robot can complete real-time image recording tasks in fully opaque pipeline networks under human teleoperation.

To highlight our contributions, this study presents a fully untethered soft in-pipe robot with tightly integrated actuation and sensing, achieving: (1) the first demonstration of untethered soft robot steering through a T-joint; (2) competitive locomotion speeds in vertical and horizontal pipelines; (3) adaptability to varying pipeline diameters; (4) proprioceptive sensing for real-time pipeline diameter mapping; and (5) integrated camera-based perception and liquid-metal contact sensing for navigation assistance in opaque environments.

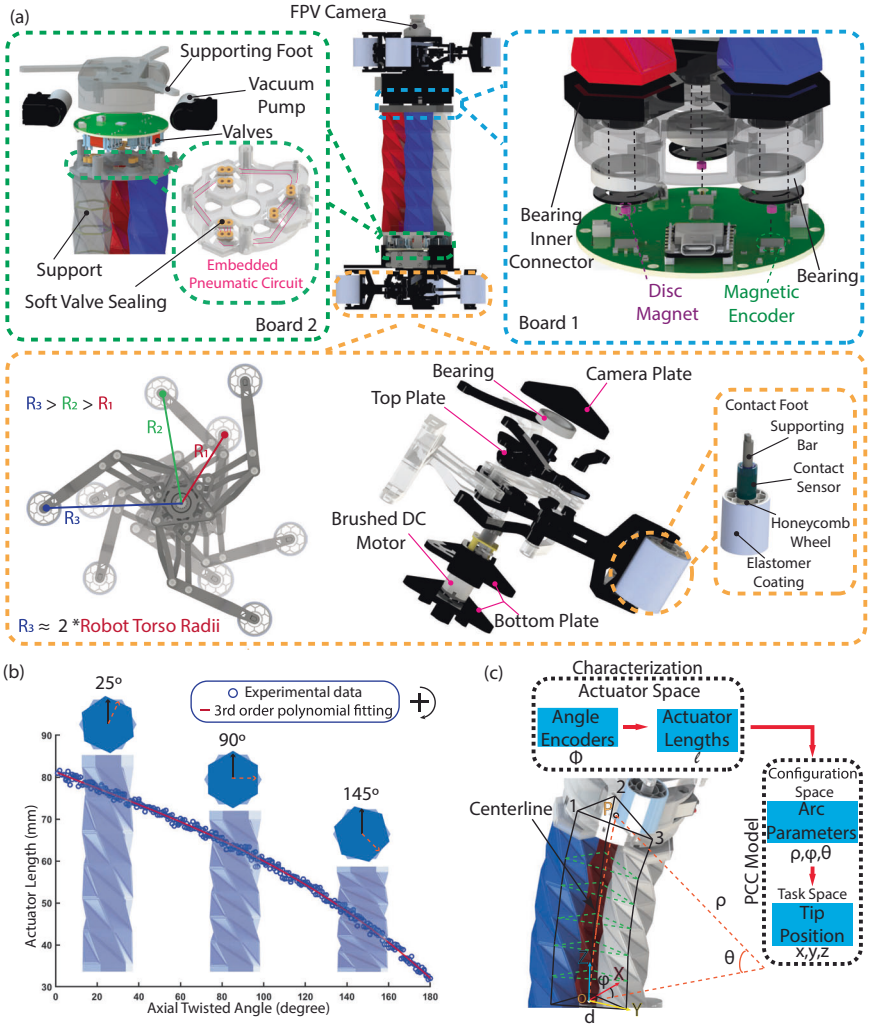


Fig. 2 | Design overview and modeling of SPPIRO. **a** Design of the propulsion and anchor modules of SPPIRO. The green box highlights the actuation end of the propulsion module with an embedded pneumatic circuit. The blue box shows the sensing end of the propulsion module. The orange box illustrates the anchor module design. **b** Characterization of the Kresling origami actuators, showing the relationship between twist angle and actuator length. **c** Kinematic modeling and spatial transformation of the robot torso using the Piecewise Constant Curvature (PCC) model.

Results

Mechanical design

SPPIRO adopts an earthworm-like peristaltic strategy: a central pneumatic propulsion section analogous to longitudinal muscles works in sequence with two anchoring modules that mimic circular muscle constrictions. The propulsion section integrates three pneumatically actuated Kresling origami elements that generate proprioceptive forward and reverse locomotion and enable omnidirectional steering. At both ends, linkage-driven anchor modules radially expand to more than twice the torso diameter, allowing SPPIRO to brace against pipes of varying diameter and to establish controlled push-pull sequences when negotiating turns and multi-directional junctions. Each anchor deploys 3D-printed, honeycomb-inspired soft contact wheels that provide radial compliance for diameter changes while maintaining axial support for load transfer during retraction and extension cycles. Liquid-metal-based contact sensors embedded in the wheels furnish contact feedback to a teleoperator, supporting confident navigation and inspection in fully enclosed pipeline networks. The design details of SPPIRO are shown in Fig. 2a and Supplementary Video S1.

Propulsion module. Inspiring from our previous work and other state of art works in Kresling origami[27–31], the propulsion module, measuring 68.5 mm in diameter and 120 mm in length, comprises three identical pneumatically driven Kresling origami actuators, fabricated via 3D printing (TPE, 75A). This module serves as a bridge connecting the sensing end (blue box) and the actuation end (green box) of the robot, as illustrated in Fig. 2a. On the actuation side, two miniature vacuum pumps and six pneumatic valves are integrated to enable untethered actuation. At this end, the valve sets on the PCB (which will be detailed in a later section) directly interface with a 3D-printed connector plate. To enhance visual neatness and simplify the system's pneumatic routing, the air channels connecting the valves and vacuum pumps are directly embedded within the 3D-printed components, as shown in Fig. 2a. Furthermore, to improve the connection and sealing between the valves and the connector plate, custom-designed orange sealing components, depicted in Fig. 2a, were 3D-printed and installed at each valve inlet and outlet. Notably, as shown in Fig. 2a, hexagonal supports are placed between each Kresling origami pattern within the actuators to prevent buckling and ensure linear contraction during vacuum operation.

At the other end of the propulsion module, referred to as the sensing end, a bearing and an associated magnet and encoder are installed on the distal end of each Kresling origami actuator. The inner ring of the bearing is connected to the inner ring connector with a disc magnet mounted, while the outer ring is attached to the robot body anchoring structure that is fixed to the anchor module. This configuration allows the bearing to decouple the rotational motion of the Kresling actuator from the anchoring structure, enabling accurate measurement of twist without twisting the robot's body.

Unlike prior studies that typically alternate the twisting direction of adjacent Kresling units to cancel out rotation during contraction[31,32], we intentionally exploit the intrinsic coupling between twisting and contraction for actuator length proprioception. The encoder records the actuator's rotational angle, which, due to the one-to-one relationship between the actuator length and twist angle (illustrated in Fig. 2b), enables robust real-

time estimation of actuator length. With all actuator lengths known, the robot's overall shape, as well as the shape of the pipeline it traverses, can be reconstructed using the Piecewise Constant Curvature (PCC) model[33]. The detailed logic behind this proprioceptive sensing approach is presented in Fig. 2c, and the specifics of the PCC model are provided in the Methods section "Kinematic model for SPPIRO and pipeline reconstruction".

Anchor module. Apart from the propulsion module, the robot consists of two identical anchor modules, measuring 41.9 mm in thickness and diameter varying from 132.8 to 209.7 mm, located at both ends. Each anchoring unit employs a single brushed DC motor coupled with a carefully designed linkage mechanism (see Fig. 2a, orange), which converts motor rotation into radial expansion. Based on kinematic analysis, this linkage mechanism achieves an expansion ratio exceeding twice the robot's torso diameter.

The anchoring mechanism design is guided by three key objectives: large expansion ratio for varying pipe diameters, robust and compact force generation, and proprioceptive capability for pipe diameter reconstruction. We evaluated several candidate approaches before arriving at the current design. Pneumatically actuated soft anchors[34] offer inherent compliance but are susceptible to puncture by debris commonly found in real-world pipelines. Traditional rigid linkages—including hinged legs[35], parallelogram mechanisms[36], and scissor-jack expanders[37]—provide structural robustness but typically achieve only 1.3–1.5 times expansion ratio and often require multiple actuators, resulting in bulky modules unsuitable for small or curved pipes. Passive spring-loaded anchors[38], while mechanically simpler, remain in constant wall contact and generate parasitic friction during peristaltic locomotion, reducing propulsion efficiency; they also produce diameter-dependent forces that degrade anchoring performance in wider pipes and cannot directly sense anchor expansion without additional displacement sensors.

Based on these considerations, we designed a motorized, compound bar linkage mechanism in which relative plate rotation is transformed into amplified radial expansion of all anchor legs. This architecture enables single-actuator synchronized deployment with expansion exceeding twice the torso diameter. The motorized actuation allows active friction modulation by retracting the moving anchor during each gait cycle, while the rigid linkage structure provides direct inference of anchor radius from motor angle through a deterministic kinematic relationship—enabling proprioceptive diameter sensing without additional sensors. Together, these features address the limitations of the alternative approaches while satisfying all three design objectives.

The contact feet, positioned at the outermost section of each anchoring module, consist of 3D-printed soft wheels (50A resin, Formlabs Inc.) featuring a honeycomb-inspired structure. This design substantially enhances the effective contact area with pipe walls due to its inherent radial compliance, enabling passive conformity to the varying inner wall geometries encountered in pipelines. Although radially compliant, the honeycomb structures maintain sufficient axial stiffness to robustly support the robot's weight, making them particularly well-suited for SPPIRO's anchoring mechanism.

To further enhance frictional contact, a silicone skin (Ecoflex 00-50, Smooth-On Inc.) is cast onto the outer surface of the soft wheels. Additionally, each honeycomb wheel contains an embedded soft contact sensor for detecting surface contact, described in detail in a subsequent subsection. Given the varying pipeline diameters, complex junction geometries, and surface irregularities such as ovality, roughness, weld beads, corrosion pits, and step transitions, achieving reliable anchoring is critical for maintaining high translational speed in peristaltic locomotion. The radially adaptive honeycomb wheels notably enhance the anchoring module's ability to maintain stable navigation and support robust loading through passive compliance with pipeline conditions.

To better understand the performance of the anchor unit and the soft wheels, we conducted a series of experiments to evaluate the anchor's load-bearing capabilities under different conditions. The experimental setup is shown in Fig. 3a. A segment of pipeline with a diameter of 145 mm was securely fastened to an Instron tensile testing machine, with a string connecting the load cell on the Instron to the top of the anchor module.

We first examined the maximum load each anchor design could withstand under various tilt angles. As shown in Fig. 3b, three candidate designs were evaluated during the robot development. The first is a 3D-printed honeycomb wheel coated with an outer layer of elastomer. The second design consists of a wheel made entirely of elastomer. The third is a pad-style anchor, whose contact surface approximates the tangent surfaces of pipelines with various diameters.

To test each design, we actuated the anchor module to grip the pipe segment and then used the Instron machine to pull the anchor while the load cell recorded the corresponding pulling force. The maximum force recorded before slippage occurred is reported as the anchor load capacity. Importantly, we tested each design at three tilt angles of 0°, 10°, and 20° to evaluate performance under misalignment—conditions commonly encountered in our real robot tests, especially during steering. Fig. 3c illustrates these three experimental configurations. To ensure a fair comparison, all anchors were fully expanded to their maximum gripping position before being pulled by the Instron machine. In the free (unloaded) state, the honeycomb wheel and the full elastomer wheel have identical nominal dimensions. All contact mechanisms were mounted on the same linkage and driven by the same motor, applying identical stall torque (12 kg cm). Consequently, any difference in effective contact diameter under maximum expansion arises from each design's intrinsic mechanical and structural properties—the honeycomb wheel deforms more due to its compliant lattice structure, while the full elastomer wheel maintains a slightly smaller compression profile—directly reflecting each mechanism's true contact mechanics under identical actuation conditions.

The experimental results are shown in Fig. 3d. Each test was conducted over five trials. Under the 0° tilt condition, the full elastomer anchor design exhibited a slightly lower load capacity, sustaining approximately 13–14N. The honeycomb wheel and pad designs showed comparable performance, each reaching around 15–16N. At a 10° tilt, both the honeycomb and pad designs experienced a reduction in load capacity, though to different extents. The honeycomb wheel showed a slight decrease to approximately 13–14N, while the pad design's performance dropped sharply to just 2–3N, barely maintaining anchoring within the pipe segment. In contrast, the full elastomer design maintained relatively consistent performance, staying between 12 and 13N. At a 20° tilt, the honeycomb wheel design achieved the highest load capacity, retaining a performance level of approximately 15–16N. Among all three designs, the honeycomb wheel also exhibited the smallest standard deviation across trials, indicating more consistent performance. These results suggest that incorporating a honeycomb structure into the anchor foot not only increases its load-bearing capacity but also enhances its adaptability and reliability in irregular pipeline conditions, such as those encountered at turning junctions.

To evaluate the generalizability of anchoring performance across different pipe diameters, we conducted additional experiments using three pipe sizes: 145, 175, and 195 mm. For each diameter, the anchor module was positioned without tilt, and its maximum anchoring load was measured following the same experimental setup. Results show that the anchor's maximum load increases with pipe diameter: from approximately 14 N in the 145 mm pipe to 16 N in the 175 mm pipe and 18 N in the 195 mm pipe (Fig. 3e). This trend arises from two geometric effects inherent to the linkage mechanism: (1) at larger expansions, the linkage bars approach a near-collinear "toggle" configuration that increases axial stiffness, and (2) the contact foot becomes more aligned with the pipe wall normal, improving force transmission efficiency. A detailed kinematic analysis of this phenomenon is provided in the Supplementary Materials section 4.

Electrical design and contact sensing

To achieve truly proprioceptive and power-autonomous operation, SPPIRO integrates onboard power, sensing, and actuation so a teleoperator can

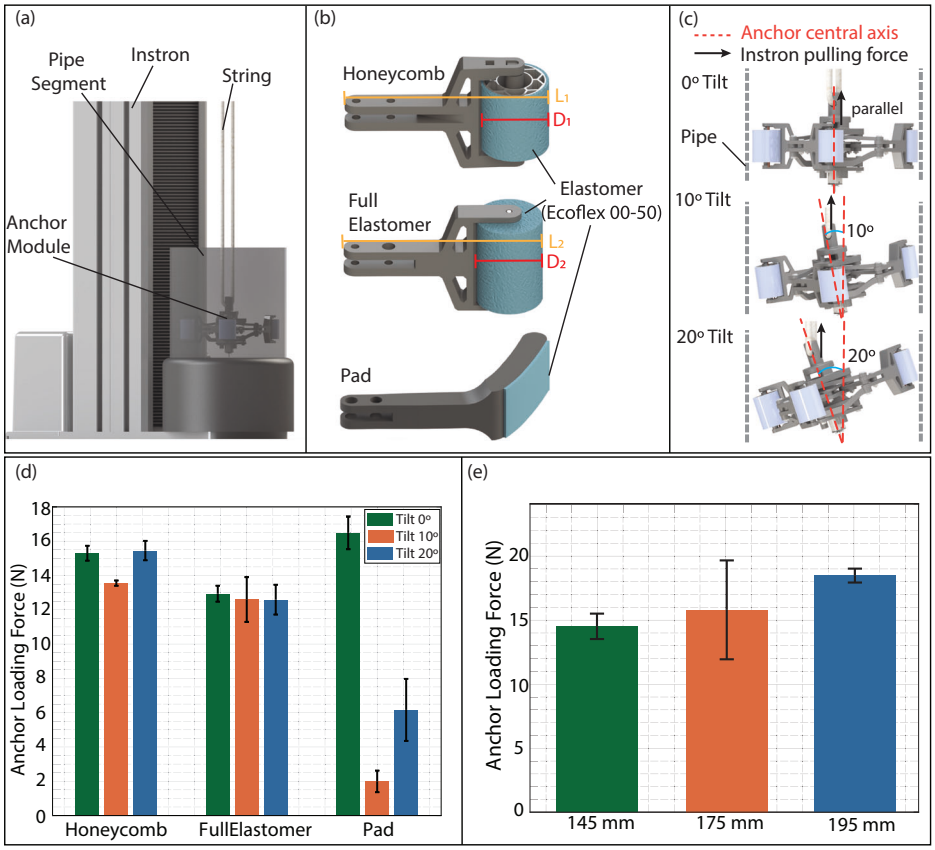


Fig. 3 | Anchor module characterization. **a** Experimental setup for anchor characterization. **b** Three contact foot design candidates. The honeycomb and full elastomer designs share the same overall length, L1 = L2 and wheel diameter, D1 = D2. All three designs are engineered to achieve identical radial expansion, enabling comparable anchoring performance in pipe segments of equivalent diameter. **c** Anchors tilted at different angles for characterization. **d** Anchor loading performance of the three contact foot designs under 0°, 10°, and 20° tilt angles. **e** Maximum anchoring load comparison across different pipe diameters (145, 175, and 195 mm). The anchor exhibits higher load capacity in a larger-diameter pipe.

remotely navigate it in challenging, opaque pipelines. The robot pairs soft contact sensing with a compact control-actuation electronics stack for reliable in-pipe operation. Each deformable wheel embeds a liquid-metal-based capacitive sensor that reports secure anchor contact through pressure-induced capacitance change, captured by an onboard module. The lightweight controller also drives the pneumatic Kresling actuators and anchor motors, while a dedicated valve-pump assembly supplies adjustable vacuum. Powered by an onboard battery, SPPIRO streams essential pose and contact data to the operator for contact-aware navigation and pipeline reconstruction.

Electrical design. SPPIRO's electrical system tightly integrates power, computation, actuation, and sensing. All electronics are powered by a 7.4 V LiPo battery, paired with a voltage regulator for appropriate voltage adjustment. A XIAO nRF52840 microcontroller (SeeedStudio Inc.) manages data transmission between the sensing and actuation modules, and also communicates with a PC via Bluetooth for teleoperation and data visualization, enabling pipeline reconstruction and contact-sensor-assisted maneuvering in opaque environments.

The microcontroller controls the three Kresling origami actuators and two anchor module motors via PWM signals. In addition to sending control commands, it also reads sensory inputs, including: (1) three angular measurements from the Kresling actuators, (2) two angular readings from the anchor motors, and (3) six capacitive measurements from the liquid-metal-based contact sensors.

The actuation module consists of two miniature vacuum pumps and three sets of solenoid valves from Parker Inc. Each set includes one 3-way valve and one 2-way valve for controlling a single actuator. The 3-way valve switches the actuator's connection between the vacuum source and the atmosphere, while the 2-way valve regulates the amount of air entering or exiting the actuator. The detailed pneumatic circuit is shown in Fig. 4a. The sensing module includes an MPR121 capacitive sensing board for reading the liquid-metal-based sensors, as well as three encoders for proprioception of the Kresling actuators. A multiplexer is used to manage I2C address switching and ensure seamless communication between modules.

All electronics are integrated into two custom PCB boards, mounted at opposite ends of the robot's torso. Board I contains the microcontroller, three-angle encoders, the multiplexer, and two bidirectional brushed motor electronic speed controllers (ESCs) for driving the anchor module motors. On the reverse side of Board I, an MPR121 capacitive sensing module is integrated, as shown in Fig. 4b. Board II houses all the actuation components, including the miniature valves and vacuum pumps. Through the experiment, SPPIRO can continuously operate for 26 minutes. A detailed energy consumption and Cost of Transport (COT) calculation for this system is shown in the Supplementary material section 3.

Liquid-metal-based contact sensor. Crucial to SPPIRO's proprioceptive navigation capability is its ability to accurately detect anchor-pipe interactions. However, teleoperation of robots becomes particularly challenging in fully enclosed pipeline environments, where visual feedback available to human operators is limited or entirely unavailable. Consequently, sensors capable of reliably detecting the quality of anchor contacts become essential. To address this challenge, we propose a liquid-metal-based capacitive pressure sensor integrated within the soft wheel (Fig. 2a, orange). This sensor enables the detection of anchor contact with pipeline walls by reading the changes in capacitance. When an anchor contacts the pipeline wall, it induces deformation in the soft wheel, consequently deforming the embedded sensor as shown in the inset of Fig. 4c. This deformation brings the two conductive layers of the sensor closer, thereby altering the sensor's capacitance. In Fig. 4c, the change in capacitance reading of a contact sensor is plotted. The anchor module is controlled to anchor and detach in a pipe segment with a diameter of

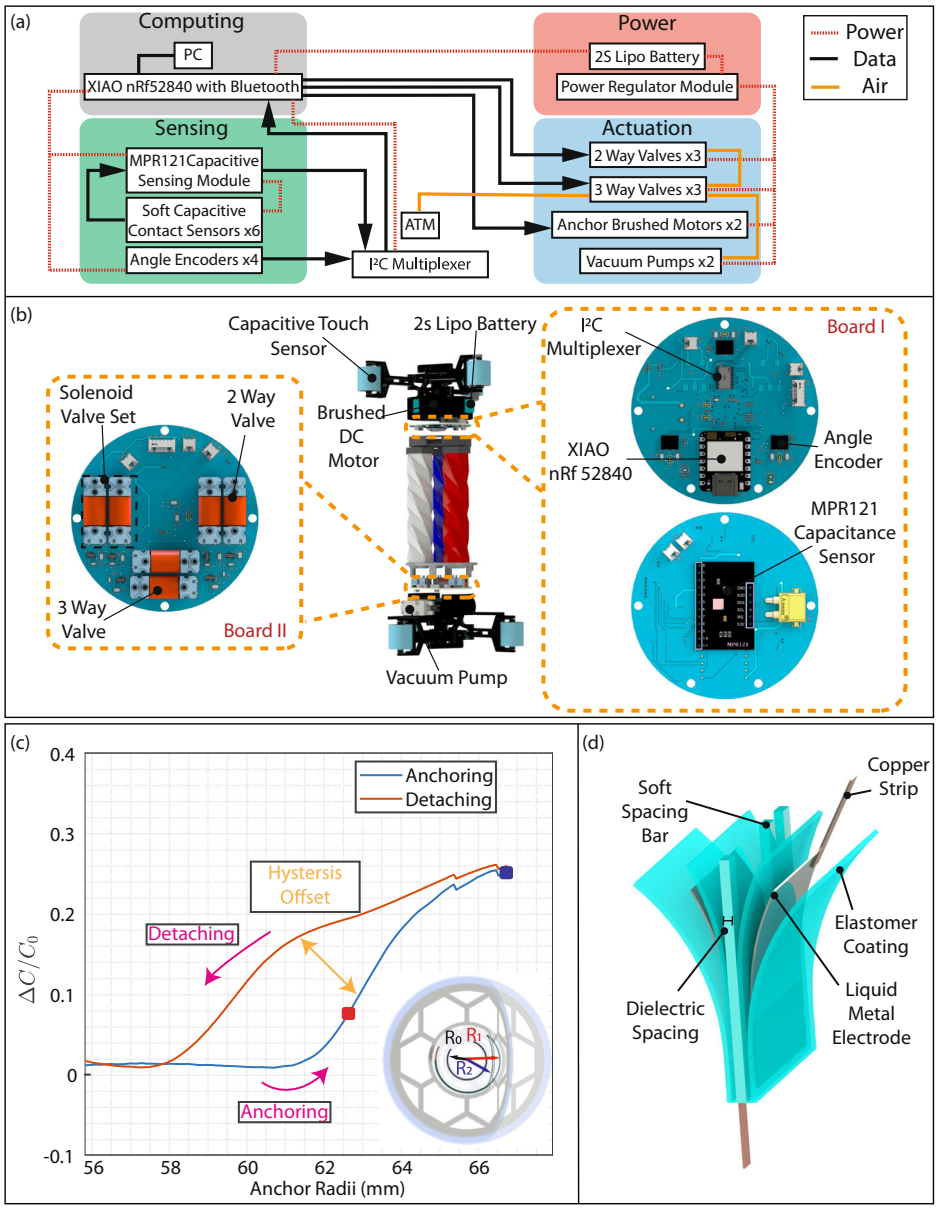


Fig. 4 | Electrical system of SPPIRO. **a** Communication architecture and mechanical/electrical connections between different modules of SPPIRO. **b** Customized PCB boards showing all IC components. **c** Capacitance readings from the liquid-metal-based contact sensor during a full anchor-detach cycle. The red squared representing the capacitance difference between contact foot state R1 and R0, and the blue square represents the difference between state R2 and R0, as shown in the figure. **d** Design of liquid-metal-based contact sensor.

145 mm. The maximum change in capacitance reading can reach about 25%, which is sufficient to determine the contact state of the anchor module. In this contact sensor, unlike conventional soft sensor designs that typically utilize elastomeric or foam-based dielectric layers, we use spacing bars to support only critical regions, as shown in Fig. 4d. These spacing bars allow the outer electrode to adopt a curved shape without requiring a continuous elastomer sheet. This design choice is motivated by the highly constrained space inside the honeycomb wheel, limiting the total sensor thickness to approximately 2 mm. Additionally, there are practical constraints on the minimum thickness of the elastomer coating around the liquid metal, since excessively thin coatings may reduce sensor lifespan due to friction against the soft honeycomb wheel during anchoring. Consequently, creating a sensitive yet easily compressible dielectric layer within these constraints is challenging. Given that variations in sensor height are the primary contributor to capacitance changes, employing spacing bars to create a largely empty dielectric region enables the sensor to more reliably detect and report subtle deformation-induced capacitance variations. The fabrication process of the sensor is detailed in the "Methods" section and Supplementary Fig. S1.

Locomotion performance of SPPIRO

Horizontal and vertical speed testing. Equipped with adaptive, sensorized, and robust anchoring mechanisms, the proposed pipeline inspection robot, SPPIRO, demonstrates outstanding locomotion performance using the gait sequence illustrated in Fig. 5a. The gait begins with the front anchor fully expanded and securely attached to the pipe wall. Subsequently, all three Kresling actuators contract to a desired length, pulling the rear anchor forward toward the front anchor. Once in position, the rear anchor fully expands to engage with the pipe wall. The front anchor then detaches, and the Kresling actuators elongate back to their original length, propelling the robot forward and completing one gait cycle.

By continuously repeating this sequence, SPPIRO performs locomotion through pipe segments of varying diameters, including 195, 175, and 145 mm, in both vertical and horizontal orientations, as shown in Fig. 5b. Each test was repeated five times, and the full locomotion process was recorded. We utilized Co-Tracker[39] to extract and analyze the robot's locomotion speed across these conditions. Due to the cyclic nature of the gait, the velocity profiles in both vertical and horizontal configurations exhibit periodic patterns (Fig. 5c).

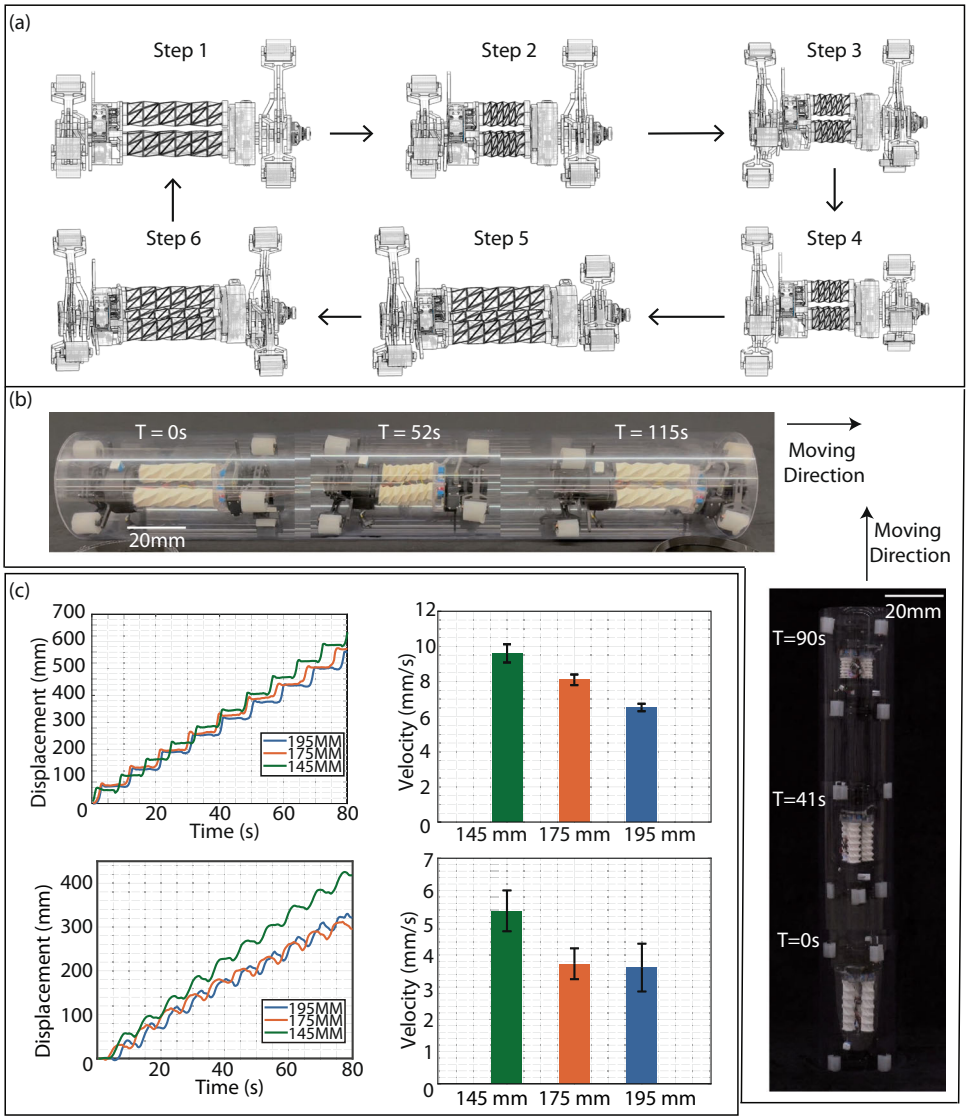


Fig. 5 | Locomotion of SPPIRO in straight pipe segments. a Locomotion gaits of SPPIRO during speed testing in a straight pipe segment. **b** Composite video frames showing SPPIRO navigating through both horizontal and vertical pipelines. **c** Locomotion speed of SPPIRO in pipelines with three different diameters, 145, 175, and 195 mm, for both vertical (top row) and horizontal (bottom row) orientations.

The experimental results are presented in Fig. 5c. The first row of the figure shows the robot's performance in vertical pipe segments. In this orientation, SPPIRO achieved its highest speed in the 145 mm diameter pipe, reaching up to 9.5 mm/s. For the 175 and 195 mm diameter pipes, the speeds were relatively similar, ranging from approximately 8.5 to 7.0 mm/s.

In the horizontal orientation, SPPIRO achieved its highest speed in the 145 mm diameter pipe, ranging from approximately 5.5 to 6.5 mm/s. The 175 and 195 mm diameter pipes exhibited similar speeds, with the 195 mm pipe reaching approximately 3.0 to 4.6 mm/s, while the 175 mm pipe showed the lowest speeds, ranging from about 3.2 to 4.2 mm/s. The second row of Fig. 5c summarizes the horizontal locomotion data.

Comparing horizontal and vertical locomotion, SPPIRO consistently achieves higher velocities when climbing vertically than when crawling horizontally across all tested pipe diameters. Specifically, the maximum speeds in the vertical orientation are approximately 1–2 mm/s greater than those observed in the horizontal configuration. This performance difference primarily arises from gravitational effects on the robot's posture during horizontal locomotion. In the horizontal setup, SPPIRO's body axis deviates considerably from being parallel to the pipe axis due to insufficient bending stiffness, which prevents the non-anchoring foot from remaining elevated. As a result, the dragging contact between the non-anchoring foot and the pipe wall introduces additional friction, hindering forward motion. Moreover, the variance in locomotion speed is noticeably greater in the horizontal orientation than in the vertical case. This increased variability arises from changes in the robot's roll angle (i.e., rotation about its longitudinal axis) during each gait cycle. Specifically, when the front anchor engages with the pipe wall, the radial expansion induces a torsional twist in the robot's torso between the front and rear anchors, leading to a change in roll angle. As a result, one of the contact feet of the trailing (dragging) anchor may periodically come into direct contact with the pipe surface, significantly increasing friction and causing a reduction in locomotion speed.

With a maximum vertical speed of 9.5 mm/s and a horizontal speed of 6.5 mm/s, SPPIRO—despite being a fully power-autonomous soft robot—achieves competitive performance compared to other tethered soft robots. Moreover, while having a size comparable to other untethered pipeline robots, SPPIRO's maximum speed, measured in body lengths per second, is generally an order of magnitude higher—approximately ten times faster. Regarding its operational pipe inner diameter range, SPPIRO can navigate through pipes with a diameter up to 1.6 times larger than the smallest diameter it can traverse, which ranks among the highest operational ranges reported. A comprehensive performance comparison with similar robots is provided in Table 1.

Complex pipeline locomotion testing. In addition to evaluating locomotion speed in horizontal and vertical pipe segments, we also

Table 1 | Performance comparison between SPPIRO and other soft in-pipe robots

Robot	Zhao[43]	Kim[47]	Liu[41]	Chen[12]	Yeh[48]	Fang[42]	Guo[49]	Liu[50]	This Work
Driving mechanism	Pneumatic	Cable	Pneumatic	Pneumatic	Cable	SMA	Pneumatic	Cable	**Pneumatic**
Max vertical speed (mm/s)	0.39	0.83	0.31	2.2	3.56	4.12	0.6	-	**9.5**
Max horizontal speed (mm/s)	1.27	0.85	0.61	2.28	5.67	-	0.8	9	**6.5**
Active steering	No	No	Yes	No	No	Yes	No	No	**Yes**
Contact sensing	No	No	No	Yes	No	No	No	No	**Yes**
Tethered/untethered	U	U	U	T	T	T	U	T	**U**
Length (mm)	1200	350	120	70	140	113	130	130	**250**
Width (mm)	82	60	38	30	34	32	38	32	**68.5**
Max speed (BL/s)	0.0010	0.0024	0.0051	0.031	0.041	0.036	0.0062	0.07	**0.031**
Operational pipe I.D. (mm)	86	65	55–85	37.2–59.6	75	35	32	155–195	**140–200**

Bold values indicate the specifications of the robot proposed in this work.

teleoperated SPPIRO to navigate more complex pipeline terrains. Specifically, we tested SPPIRO's ability to traverse a 90° sharp turn in a 145 mm diameter pipeline in under 94 s, as shown in Fig. 6a and Supplementary Video S2. By leveraging differential actuation of the three Kresling origami actuators, SPPIRO achieves omnidirectional bending, while the elastomer-coated honeycomb wheels provide secure anchoring at various tilt angles.

Through repetition of the bend-anchor-advance sequence, SPPIRO successfully navigated the T-junction, completing a 90° turn from vertical to horizontal orientation within 166 s. Notably, during such turning maneuvers, the front anchor requires at least two feet to make contact with the pipe wall to stably support the robot's weight. If only one foot maintains contact, the anchor lacks sufficient friction to hold the robot, resulting in continuous slipping or falling (see Fig. 6b, second row right, and Supplementary Video S3). Through repeated trial and error, SPPIRO eventually spun around its longitudinal axis, due to the same roll-angle-induced torsion described in Section "Horizontal and vertical speed testing", to reach a configuration where at least two feet were in contact with the pipe wall, allowing it to regain stable anchoring. This anchor adjustment process is demonstrated in Supplementary Video S3.

This behavior stems from the inherent characteristics of the three-foot anchoring design, which can lead to momentary contact loss and motion singularities. However, compared to traditional rigid in-pipe robots that rely on continuous contact from tracks or multiple wheels[40], SPPIRO maintains relatively stable anchoring even in such singular configurations—highlighting the advantages of its compliant, adaptive design.

Equipped with adaptive anchor modules and a proprioceptive propulsion torso, SPPIRO is capable of navigating pipelines with varying diameters and reconstructing their dimensions. As shown in Fig. 6c, SPPIRO successfully maneuvered through three interconnected pipelines with diameters of 145, 175, and 195 mm, and a total length of 1050 mm. The reconstructed pipeline geometry during locomotion is visualized in Fig. 6c and Supplementary Video S6. The entire process was completed in under 400 s. This extended duration was primarily due to the limited bandwidth of Bluetooth Low Energy (BLE), which slowed down the data transmission required for pipeline reconstruction.

In addition, we also conducted a preliminary demonstration of SPPIRO operating within an opaque pipeline environment for defect detection. Contact sensors were used to assist teleoperation, while an onboard camera enabled video documentation and visual identification of defects. This demonstration highlights SPPIRO's practical capabilities in real-world conditions, as shown in Fig. 6d and Supplementary Video S7.

For practical applications, to evaluate the robustness of SPPIRO under realistic pipeline conditions, we conducted additional locomotion experiments on surfaces beyond smooth PVC. Specifically, we tested the robot's performance on (i) wet and lubricated PVC surfaces, (ii) clean metal pipes, and (iii) metal pipes coated with a thin layer of mud to simulate contaminated industrial environments. These experiments were conducted in vertical pipes with a diameter of 145 mm, and the results are compared against the clean PVC baseline under identical conditions.

As shown in Fig. 6e, SPPIRO achieves comparable locomotion speeds across all four surface conditions. The robot attains an average velocity of approximately 9.5 mm/s in clean PVC pipes, 10.1 mm/s in wet PVC pipes, 8.5 mm/s in clean metal pipes, and 6.9 mm/s in soiled and rusted metal pipes. While there is a slight decrease in velocity under more challenging surface conditions, the overall locomotion performance remains consistent and robust.

This behavior can be attributed to the design of the honeycomb-inspired soft contact wheels and the mechanics of peristaltic locomotion. First, the velocities on clean PVC and wet PVC are nearly identical, indicating that moisture does not substantially affect locomotion performance. This is because the motorized anchor module generates sufficient preload force to overcome surface slipperiness—the compliant honeycomb structure conforms to the pipe wall and maintains a stable grip even under wet conditions.

On clean metal surfaces, the locomotion speed decreases slightly to approximately 8.5 mm/s. This reduction can be attributed to the higher friction coefficient between the elastomer contact feet and the metal surface compared to the elastomer-PVC interface. During peristaltic locomotion, the non-anchoring foot is not strictly vertical but rather tilted due to the robot's body posture. This tilted anchor still contacts the pipe wall with residual normal force, and the higher elastomer-metal friction generates greater resistance during the translation phase. As a result, more energy is dissipated in overcoming this parasitic friction, leading to a slight reduction in net forward velocity.

The most notable velocity reduction occurs on soiled and rusted metal pipes. We attribute this primarily to rolling friction introduced by small pebbles and debris embedded in the mud layer. When the soft contact feet press against the contaminated surface, these loose particles become trapped at the contact interface. Rather than remaining stationary under shear loading—as would occur on a clean surface—the particles rotate in place, acting as miniature rolling elements between the elastomer and the pipe wall. This rolling motion requires far less tangential force to initiate than static friction, effectively lowering the friction coefficient at the anchoring interface. As a result, some of the force that would otherwise propel the robot forward is lost to micro-slip events at the anchored foot, reducing the effective grip during the pushing/pulling phases and resulting in reduced net forward displacement per gait cycle.

These results demonstrate that SPPIRO maintains stable anchoring and consistent forward locomotion across varied and challenging surface conditions, validating the effectiveness of the honeycomb wheel design and motorized anchoring mechanism for real-world pipeline inspection applications. Detailed process is shown in Supplementary Video S9.

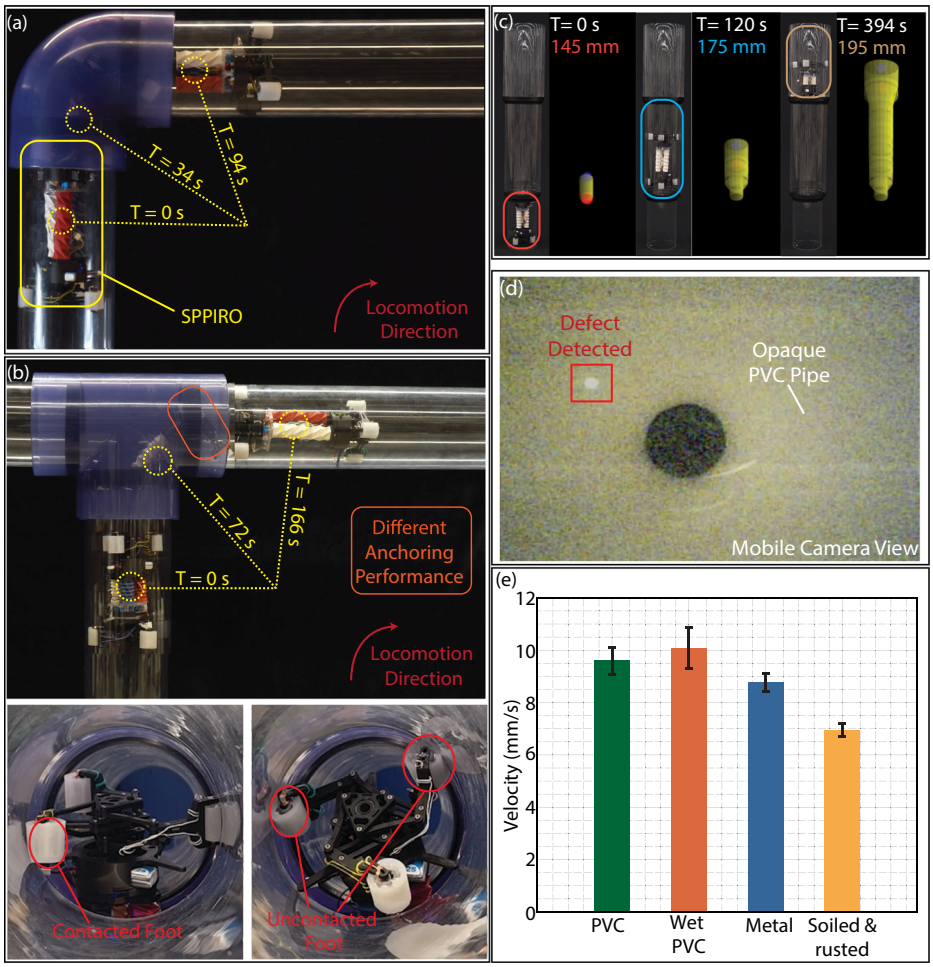


Fig. 6 | Locomotion performance of SPPIRO in complex pipeline environments. a SPPIRO navigates through a confined 90° sharp turn. **b** SPPIRO selects a direction and maneuvers through a T-junction. The one-foot and two-foot contact states illustrate motion singularities during T-junction navigation. **c** SPPIRO traverses a varying-diameter pipeline consisting of interconnected segments with diameters of 145, 175, and 195 mm. The pipeline geometry is reconstructed alongside the robot's locomotion. **d** SPPIRO performs an inspection task in a simulated opaque PVC pipeline. **e** Locomotion performance of SPPIRO on different pipe surface conditions. Velocity comparison among four different pipe surface media: clear PVC (green), wet PVC (orange), clean metal (blue), and soiled/rusted metal (yellow). SPPIRO maintains consistent locomotion performance across all surface conditions, with only minor velocity reductions observed on rougher and contaminated surfaces.

Discussion

In this paper, we introduced SPPIRO, a soft, power-autonomous, proprioceptive in-pipe robot designed for adaptive inspection in complex pipeline environments. SPPIRO demonstrates high-performance, fully untethered locomotion across horizontal and vertical pipelines of varying diameters. Quantitatively, SPPIRO achieves a maximum vertical speed of 9.5 mm/s and horizontal speed of 6.5 mm/s, comparable to state-of-the-art tethered in-pipe robots[11,12,41,42] and substantially faster than previously reported untethered pneumatic soft pipeline robots[43]. Mechanically, SPPIRO integrates three parallel 3D-printed Kresling origami actuators, enabling large contraction (70%), omnidirectional steering, and 3D mobility. This configuration allows the robot to actively navigate multidirectional joints. Beyond actuation, we leverage the embodied mechanical intelligence of the Kresling origami—specifically, the strong coupling between axial twisting and contraction length—together with onboard encoders, to enable SPPIRO to estimate its shape and reconstruct pipeline geometry using the Piecewise Constant Curvature (PCC) model. Furthermore, the inclusion of liquid-metal-based soft contact sensors, combined with an onboard camera, equips SPPIRO with the ability to proprioceptively detect anchoring states and perform visual inspection in fully opaque pipeline environments. This capability was demonstrated through successful defect detection during teleoperated navigation in a simulated pipeline.

While SPPIRO demonstrates strong locomotion capabilities in various pipeline environments, several limitations in the current design must be addressed to enable real-world deployment. First, the three-foot anchoring module introduces motion singularities during turning maneuvers. Although the spiral configuration allows continuous roll adjustments to partially compensate for this, the process remains inefficient, requiring considerable time for reorientation. Second, when all three Kresling origami actuators are vacuumed to approximately half their length, the significant reduction in bending stiffness leads to sagging of the anchoring end—particularly in horizontal pipelines. This sag increases friction with the pipe wall and reduces actuator efficiency during pushing or pulling, thereby limiting overall locomotion speed. Third, the pipeline reconstruction system currently relies solely on the Piecewise Constant Curvature (PCC) model, which assumes smooth curvature along the robot's body. This assumption holds in relatively open or vertical segments, but frequent contact with the pipe wall—especially during horizontal travel and turning—violates the model, reducing reconstruction accuracy.

To overcome these challenges and prepare SPPIRO for real-world deployment, several improvements are planned. First, increasing the bending stiffness of the torso could help prevent unintentional contact between the non-anchoring feet and the pipe wall during horizontal locomotion, thereby improving speed and efficiency. However, this comes with a trade-off: stiffer Kresling actuators may require longer actuation and vacuum durations, necessitating design optimization. An alternative solution involves adding a second robot segment—shortened and connected in series—to provide additional support and stability. This segmented configuration would improve locomotion reliability by introducing an intermediate anchoring point, reducing drag, and simplifying control of the moving anchor. Second, enhancing the contact sensing system with more sensitive and responsive sensors would enable smarter anchoring strategies. This would allow the robot to apply optimal pressure against the pipe wall, reducing unnecessary motor load, conserving energy, and extending the lifespan of the anchoring motors by avoiding operation under stall torque. Finally, integrating onboard state estimation, SLAM, and path planning

capabilities is essential for achieving full autonomy, particularly in steering and navigating complex junctions. Together, these enhancements would pave the way for SPPIRO to evolve into a robust, fully autonomous pipeline inspection platform ready for deployment in real-world scenarios.

Methods

Fabrication of Kresling origami actuators

We present a 3D printing approach for fabricating Kresling origami structures using a commercially available fused deposition modeling (FDM) printer (X1C, Bambu Lab) as illustrated in the Supplementary Fig. S8. In this method, thermoplastic elastomer (TPE, SpiderMaker, 75A) is utilized to form the flexible body of the actuator, while polyphthalamide with carbon fiber (PPA-CF, Bambu Lab) is employed to create the rigid components. One of the primary challenges associated with FDM technology is ensuring the airtightness of printed components. To address this, we implemented targeted optimizations—such as fine-tuning the retraction parameters and adjusting print speed—which are summarized in the Supplementary Table S1. During the slicing process, the origami chamber is oriented vertically, and the support angle is increased to minimize the formation of support structures. This adjustment effectively preserves the integrity of the chamber's surface, preventing potential damage caused by support removal.

Fabrication of a liquid-metal-based contact sensor

The contact sensor fabrication process can be separated into two procedures: the electrode stage and the spacing bar stage. In the electrode fabrication stage, we prepared a 1:1 ratio of the two-part Ecoflex 00-50 silicone rubber compound and evenly coated it onto an acrylic plate using a spin coater at 300 rpm for 60 s. The elastomer was subsequently cured at 50 °C. Once fully cured, we attached a thin PET mask to the elastomer surface and precisely outlined a square shape (width: 18.4 mm, height: 22 mm) using a laser cutter. We then removed the cut PET square and brushed a thin layer of OGaIn onto the exposed area. After removing the remaining PET mask, we attached a copper strip at one corner to provide electrical contact. Finally, we spun-coated another layer of the Ecoflex 00-50 silicone elastomer, mixed at the same 1:1 ratio, on top at the same speed and duration, completing the electrode once cured. For the outer electrode layer, we made the OGaIn-coated area slightly larger (approximately 20 mm wider) than the inner electrode to ensure a gap between the inner and outer electrodes.

The spacing bar fabrication began by creating channel patterns on an acrylic plate using a laser cutter. We poured a 1:1 mixture of Ecoflex 00-50 silicone elastomer over the patterned acrylic plate and evenly distributed it using a spin coater at 300 rpm for 30 s to fill the channels uniformly. Once cured, we carefully peeled off the silicone and cut the spacing bar to the desired dimensions for assembly.

For final assembly, we used Sil-Poxy adhesive (Smooth-On Inc.) to bond the sensor components together and attached the completed sensor assembly onto a rigid support bar in the center. The whole process is presented in Supplementary Fig. S1.

Kinematic model for SPPIRO and pipeline reconstruction

The kinematic model used by SPPIRO for pipeline reconstruction is the Piecewise Constant Curvature (PCC) model. Widely adopted in soft continuum robotics for state estimation, the PCC model offers a balance of simplicity and effectiveness[27,33,44–46]. In SPPIRO, each segment of the Kresling origami structure is assumed to exhibit constant curvature. This model establishes a mapping between the actuation space, configuration space, and task space of the robot, thereby enabling pipeline shape reconstruction. The underlying logic is illustrated in Fig. 2c. As shown in the figure, the base coordinate frame $o-xyz$ is located at the center of the base. The actuator lengths ℓ in the actuation space are represented by the vector:

$$l = [l_1, l_2, l_3] \tag{1}$$

The configuration space of SPPIRO is defined by arc parameters: the radius ρ, the bending angle θ, and the orientation angle φ of the bending plane in the base frame. These arc parameters describe the curvature of the robot's centerline. The actuator lengths l are obtained using the characterized mapping shown in Fig. 2b and serve as inputs to the PCC model.

The model begins by assigning azimuthal positions to the three actuators, spaced 120° apart, as defined in Eq. (2). Based on the differences in actuator lengths, a dimensionless ratio encoding the direction of the bending plane is computed (Eq. (3)). Using these ratios, the arc parameters φ and ρ are then calculated via Eqs (4) and (5), respectively. Subsequently, a scaling factor for the arc length, based on the geometric model, is introduced in Eq. (6), which is then used to compute the bending angle θ (Eq. (7)).

Finally, with all arc parameters determined, the position of the robot's end-effector (i.e., the tip of the front anchor module) in Cartesian coordinates is computed using Eqs (8), (9), and (10). In these equations, the constant e accounts for the additional length from the tip of the propulsion module (point P in Fig. 2c) to the tip of the front anchor module.

$$\alpha_1 = \frac{\pi}{2},\ \alpha_2 = \alpha_1 + \frac{2\pi}{3},\ \alpha_3 = \alpha_1 + \frac{4\pi}{3}. \tag{2}$$

$$K = \frac{L_1 - L_2}{L_1 - L_3} \tag{3}$$

$$\varphi = \tan^{-1}\left(\frac{K+1}{\sqrt{3}\,(K-1)}\right) \tag{4}$$

$$\rho = \frac{r\big(L_1\cos(\alpha_2 - \phi) - L_2\cos(\alpha_1 - \varphi)\big)}{L_1 - L_2} \tag{5}$$

$$D = 2\,c_n\big(\rho - r\cos(\alpha_1 - \varphi)\big) \tag{6}$$

$$\theta = 2\,c_n \sin^{-1}\left(\frac{L_1}{D}\right) \tag{7}$$

$$x = x_1 + e\sin\theta\cos\varphi \tag{8}$$

$$y = y_1 + e\sin\theta\sin\varphi \tag{9}$$

$$z = z_1 + e\cos\varphi \tag{10}$$

Data availability

The corresponding data, including the anchor loading performance tests, straight-pipe locomotion tests, and the contact sensor capacitance reading, are included in the Github repository : https://github.com/zexiongc/SPPIRO.

Code availability

The code to reproduce the pipe reconstruction using PCC model, contact sensing, as well as running the whole system is published in github: https://github.com/zexiongc/SPPIRO.

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Author contributions

Z.C., X.Z., J.W. and X.H. designed the robot. Z.C., X.Z. and J.W. conducted the experiments. Z.C., X.Z., J.W. and X.H. analyzed the data. Z.C., X.Z. and X.H. wrote the main manuscript. All authors reviewed the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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